

Adaptive Network Traffic Anomaly Classification

Comprehensive Capstone Project & Technical Evaluation Report

Author: Anil M

GitHub Repository: [Anil-192005/adaptive-network-traffic-anomaly-classification](https://github.com/Anil-192005/adaptive-network-traffic-anomaly-classification)

Core Technology: Python 3.12, Scikit-Learn, FastAPI, Streamlit, Cloudflare Tunnel

Target Application: Real-time Enterprise Network Intrusion & Threat Detection

1. Executive Summary

This capstone project presents an end-to-end, zero-cost machine learning software application designed to classify computer network traffic flows as either **Benign (Normal)** or **Anomalous (Malicious)** in real time. Rather than relying on computationally heavy, black-box deep learning architectures, the system harnesses foundational machine learning algorithms (Logistic Regression, K-Nearest Neighbors, and Decision Trees) combined with leakage-free preprocessing pipelines. The resulting solution is deployed as a fully decoupled web application featuring an asynchronous REST API backend built on **FastAPI** and an interactive cyber threat dashboard built on **Streamlit**.

2. Problem Statement & Objectives

Modern computer networks face continuous multi-vector threats including port scanning, Distributed Denial-of-Service (DDoS) floods, and covert data exfiltration. Conventional signature-based intrusion detection engines fail when confronted with polymorphic payloads and novel zero-day attacks. Furthermore, full packet inspection violates user privacy and is computationally intractable over high-speed encrypted channels.

Primary Project Objectives:

- Develop a **zero-cost, payload-agnostic** classifier operating strictly on statistical flow telemetry (durations, packet counts, byte volumes, and flags).
- Enforce **leakage-free preprocessing** where scaling and categorical encoding parameters are learned strictly from training data.
- Serve predictions through an enterprise-grade, high-throughput **REST API (FastAPI)** decoupled from the user interface.
- Provide an accessible, intuitive **web interface (Streamlit)** supporting manual flow inspection, preset scenario testing, and risk stratification.
- Expose the system with **global live URLs** accessible across mobile, tablet, and desktop devices without proprietary subscription fees.

3. System Architecture



- ColumnTransformer (StandardScaler + OneHotEncoder)

- Champion Classifier (Regularized Logistic Reg)

- Risk Tier Engine: LOW | MEDIUM | HIGH | CRITICAL

4. Dataset Specifications & Feature Schema

The model processes 8 core network flow telemetry features modeled after canonical intrusion datasets (CIC-IDS2017 & NSL-KDD):

Feature Name	Type	Description	Typical Benign Range	Attack Signature Behavior
flow_duration	Float	Connection lifespan (seconds)	1.0s – 60.0s	< 0.1s (Port scan) or high (Exfil)
packet_count	Integer	Total packets in conversation	10 – 300	1–3 (Scans) or > 1,000 (SYN Flood)
byte_count	Integer	Total bytes exchanged	1,000 – 250,000	< 100 bytes (Scans) or > 5MB (Exfil)
packet_rate	Float	Packets per second	5 – 50 pkts/s	> 500 pkts/s (DoS/Flood)
byte_rate	Float	Bytes per second	1 KB/s – 100 KB/s	Massive throughput or micro-bursts
protocol_type	Categorical	Transport protocol	TCP, UDP, ICMP	UDP floods, ICMP echo attacks
service	Categorical	Application layer service	HTTPS, HTTP, DNS	Unassigned port attempts, SSH probes

Feature Name	Type	Description	Typical Benign Range	Attack Signature Behavior
flag	Categorical	Connection state flag	SF (Normal Finish)	S0 (SYN no ACK), REJ (Rejected)

5. Model Development & Experimental Evaluation

Three foundational machine learning models were trained and benchmarked against an unseen, stratified test split of 1,201 flows (841 Benign, 360 Anomalous):

Candidate Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Status
Logistic Regression (Balanced)	100.0%	1.0000	1.0000	1.0000	1.0000	Selected Champion
K-Nearest Neighbors (k=5, Distance)	100.0%	1.0000	1.0000	1.0000	1.0000	Candidate
Decision Tree (Max Depth=8)	99.92%	0.9972	1.0000	0.9986	0.9994	Candidate

Confusion Matrix (Champion Model - Test Set):

	Predicted Normal (0)	Predicted Anomaly (1)
Actual Normal (0)	841 (True Negatives)	0 (False Positives)
Actual Anomaly (1)	0 (False Negatives)	360 (True Positives)

6. Production Deployment & Live Access

The application has been deployed across multiple tiers:

- **GitHub Repository:** <https://github.com/Anil-192005/adaptive-network-traffic-anomaly-classification>
- **Live Web Application (Accessible on Mobile & Desktop):** <https://nikon-module-literacy-weather.trycloudflare.com>
- **Live REST API Swagger Explorer:** <https://stands-eligible-incentive-cells.trycloudflare.com/docs>

7. Conclusion

The Adaptive Network Traffic Anomaly Classification project successfully illustrates that robust, low-latency, interpretable intrusion detection does not require heavy deep-learning frameworks or paid commercial APIs. By pairing leakage-free preprocessing with foundational models and an asynchronous microservices architecture, the project provides a deterministic, zero-cost cyber defense capability suitable for production network telemetry.